



Examination	University	Department	Year	CGPA
Post Graduation	IIT Bombay	CSRE	2027	9.18
Graduation	Gujarat Technological University	Information Technology	2024	8.88
Diploma	Gujarat Technological University	Information Technology	2021	10.00

WORK EXPERIENCE

- **AI Engineer (Freelance)** | Dhi Labs AI, Ledgerline; Outlyn (GATE CS & DA ed-tech), team of 4 (Apr'26-Present)
 - Engineered **Ledgerline's multi-phase LLM agent** (Discovery, ReAct and Synthesis) answering covenant questions over **350+ page** credit agreements via **23** document tools, capped at **15** iterations with full tool-call tracing for cost, latency and stop reason.
 - Cut client LLM spend per **350-page** agreement (extraction **\$10 to \$1-2**, end-to-end answering **\$5-6 to \$2**) by right-sizing the model per task and caching prompts and artifacts, with **no accuracy regression** on the expert-authored ground-truth benchmark set.
 - Built a **deterministic preprocessing pipeline** (8 LLM stages) indexing raw PDFs into TOC, definitions and cross-references, so the agent reasons over **document structure, not raw text chunks**, with deterministic reruns of the same question across releases.
 - Implemented **multi-provider LLM routing** (OpenAI / Anthropic / Grok) in YAML with prompt caching, automatic failover between providers and **Langfuse** cost and latency tracing per agent run, so provider swaps stay config-only, never code.
 - Built an offline **LLM-as-judge evaluation harness** scoring agent answers against **10** expert-authored ground-truth Q&A workbooks on completeness and accuracy, used as a regression gate before any prompt or model change ships to production.
 - Delivered **precomputed, pixel-accurate citation highlighting**, where every cited section resolves to page plus bounding boxes, reducing modal cold-open from approximately **8s** using progressive resolution, a shared artifact cache and runtime warmup, so citations open instantly for the reviewer instead of being recomputed on every request, which kept the modal responsive.
 - Exposed the platform over the **Model Context Protocol** (38 tools, 6 resources, 4 prompts) and owned the multi-tenant FastAPI / PostgreSQL / Redis backend behind it, with a **SKIP LOCKED** worker pool, advisory-lock watchdog and **528** passing tests.
 - Built **Outlyn's AI study-coach agent** that reads a student's progress, plans tomorrow's study and saves it in one database transaction; pre-loading the data it asked for most nights cut its model calls from **about 5 a night to about 2** and its cost by more than half.
 - Added **fallback across LLM providers** (Anthropic, OpenRouter), pausing a model that keeps failing for a few minutes and moving its requests to a backup model, with every switch logged, so the app keeps working when one provider is down or slow to respond.
 - Tech:** Python, FastAPI, PostgreSQL, Redis, S3, LLM APIs (OpenAI/Anthropic/Grok), ReAct Agents, MCP, RAG, Langfuse, Prompt Engineering, Pydantic, Docker
- **Project Intern** | Space Applications Centre (SAC), ISRO, Ahmedabad (May'26-Jul'26)
 - Contributed to a satellite image mosaicking workflow stitching overlapping high-resolution scenes into a single geometrically corrected mosaic by refining **RPC sensor models** from tie-point measurements in a reproducible processing pipeline run end to end.
 - Worked on the image-matching stage, evaluating **deep-learning-based image matching** alongside classical feature-based methods (SIFT/SURF) and comparing the geometric quality achieved by each approach across the same overlapping scenes.
 - Contributed to product generation and review tooling (**Cloud Optimized GeoTIFF** output, quick-look overlays, GIS footprints and geometric quality reporting), including a browser-based interface for inspecting matching results without desktop GIS.
 - Tech:** Python, PyTorch (TorchScript), OpenCV, GDAL, Rasterio, NumPy, FastAPI, React, RANSAC, Homography Estimation, RPC Sensor Models, COG, Photogrammetry, Remote Sensing
- **Freelance Software Developer** | Self-Employed (Jun'24-Jun'25)
 - Delivered production backend systems for **retail clients**, owning requirements through schema design, implementation and deployment, and handing each system over with operating documentation the client could run against without me.
 - Implemented JWT authentication and role-based access control across multiple user tiers, plus inventory, billing and payment workflows on MongoDB and PostgreSQL, with third-party API integrations for payments, messaging and invoicing.
 - Tech:** Node.js, Express.js, MongoDB, Mongoose, PostgreSQL, Redis, REST APIs, JWT, RBAC, Git, Linux
- **Software Engineer Intern** | RapidOps Inc. (Jan'24-May'24)
 - Implemented backend APIs for production-scale applications using Node.js and REST service patterns under load.
 - Optimised database queries and API flows by profiling slow paths and refactoring data access to improve response time.
 - Tech:** Node.js, Express.js, MongoDB, MySQL, REST APIs
- **Software Engineer Intern** | HumBee Studio (Feb'23-May'23)
 - Implemented backend modules and features for web applications across the request-handling and data layers.
 - Tech:** JavaScript, Node.js, REST APIs

KEY PROJECTS

- **Multi-Agentic AI for Critical Events Monitoring & Management** [Code] | Seminar Project, GNR-694, IIT Bombay (Jan'26-May'26)
 - Designed a **hierarchical multi-agent architecture** in which a Coordinator agent decomposes an earthquake scenario into specialist sub-tasks across Infrastructure and Transportation LLM agents, then fuses their structured JSON reports into a single prioritised emergency-response decision with an explicit confidence level and traceable evidence behind every recommendation it makes.
 - Grounded each agent with a curated **domain knowledge base** (6 topic categories, approx. **50** seismic and urban-vulnerability facts) injected into prompts, reducing LLM hallucination and making every risk classification traceable and auditable.
 - Enforced **schema-constrained JSON** outputs with parsing and fallback error handling over an Express REST API and CLI; design grounded in a comparative survey of **CAMEL, AutoGen, ReAct and Toolformer** agent frameworks and orchestration patterns.
 - Exposed the plan over a REST API and a CLI runner, so scenarios can be replayed, diffed and compared across model and prompt versions without re-running the agents by hand, making evaluation repeatable across releases and model upgrades.
 - Tech:** Node.js, Express.js, Google Gemini API, Multi-Agent Orchestration, Prompt Engineering, Structured Outputs, REST APIs
- **SETU: Spec-Driven Agentic Extraction of Scientific Data** | M.Tech Thesis under Prof. Pritam Das (2026-Present)
 - Building **SETU**, an **agentic LLM pipeline** that extracts structured data from scientific papers into a **provenance-tracked dataset**, storing the page and quote behind every value; a config-driven extraction spec keeps domain, format and model out of the code.
 - Designed an **escalation ladder** using **self-consistency** (2 of 3 samples must agree), then a **tool-calling agent loop** capped at **5** turns, then **Qwen3-235B**, and finally **human-in-the-loop** review, so the expensive 235B model only runs on the hard cases.
 - Added **three validation gates** before any extracted value is accepted, covering **JSON-schema** checks on the structured output, **unit and dimension** checks with pint, and **grounding** that requires the quote to appear verbatim on the cited source page.

- Tech:** Python, LLM Agents, Tool Calling, Structured Outputs (guided JSON), RAG, Hybrid Retrieval, pint, pydantic, SQLite FTS5, vLLM
- **Landsat Land-Cover Classification & Water-Body Change Detection** [Code] | *Self-Project* (Nov'25)
 - Implemented **K-Means from first principles in NumPy**, with vectorised centroid assignment, empty-cluster reinitialisation and L2-shift convergence, over approx. **1 million** pixel feature vectors, under an explicit no-scikit-learn constraint throughout.
 - Engineered physically interpretable features (**NDVI, NDWI, brightness**) and a centroid-statistics heuristic solving the cluster-to-class correspondence problem, making output deterministic under a fixed seed and quantifying water-body change across **2005–2025** directly from the classified maps, without any labelled training data or manual thresholding, all under a fixed random seed.
 - Validated cluster quality against ground-truth land-cover samples and produced change maps quantifying water-body loss and gain across the study period, with per-class accuracy reported for spectrally similar cover types such as wet soil and shadow.**Tech:** Python, NumPy, Rasterio, Matplotlib, Tkinter, Unsupervised Learning, Remote Sensing
 - **GIS Ambulance Routing & Emergency Response Analytics** [Code] | *Self-Project* (Oct'25-Dec'25)
 - Built a full-stack Flask + Leaflet application logging, routing and analysing campus ambulance trips across **18+** geocoded pickup points, backed by a SQLAlchemy schema over SQLite and Leaflet on an OpenStreetMap basemap for the operator view.
 - Added a **3-tier routing fallback** holding distance and ETA computation under API-key exhaustion, plus **isochrone service-area zones** (3/5/7-minute) and a route-frequency heat map from trip-segment aggregation, exposed as interactive overlays.
 - Added odometer tracking, trip logging and CSV export, replacing manual record-keeping for response-time reporting.**Tech:** Python, Flask, SQLAlchemy, SQLite, Leaflet.js, OpenStreetMap, OpenRouteService, OSRM
 - **TruckIt: Real-Time Logistics Booking & Fleet Tracking** [Code] | *Course Project, GNR 605: Principles of GIS* (Oct'25-Dec'25)
 - Built a two-sided freight booking system exposing **20+ REST endpoints** via Flask Blueprints over a normalised 4-table schema, with live map tracking, multi-stop planning and separate shipper and driver portals with server-side validation on each route.
 - Routed on the OpenRouteService **HGV profile** behind a three-tier **ORS to OSRM to Haversine** fallback, with a thread-safe live-tracking simulator (`threading.local`, SQLite WAL) keeping two concurrent clients in sync in real time without a broker.
 - Implemented dynamic pricing over distance, cargo weight, fleet mix and peak-hour multipliers to **1.25x** and prorated halt charges, so every quote is reproducible from the booking parameters alone, with no manual override and no hidden surcharges.**Tech:** Python, Flask, SQLite, REST APIs, Leaflet.js, OpenRouteService, OSRM, Multithreading
 - **StreamScribe: Real-Time Scene Captioning over WebRTC** | *Self-Project* (Jun'26-Aug'26)
 - Built a **3-stage real-time video pipeline** ingesting webcam/screen-share over **WebRTC** (aiortc) and streaming natural-language captions back to the browser over WebSockets at **2–3s end-to-end latency** under sustained streaming on a single GPU.
 - Cut vision-language model inference calls **60–70%** with a two-tier deduplication gate: track-ID on **YOLOv8 + ByteTrack**, then cosine similarity (threshold **0.985**) on pooled embeddings, cutting inference cost and latency on every captioning session.
 - Designed an event-driven architecture with **bounded thread-safe queues** between stages, giving natural backpressure and drop-frames-over-lag behaviour so a slow GPU degrades caption rate instead of stalling the ingest loop or dropping frames silently.
 - Generated post-session narratives via a **MAP-REDUCE LLM orchestrator** with event compression and chunking; only text ever leaves the machine, never raw video frames, giving **privacy by design** on the user's own machine by construction, not by policy.**Tech:** Python, FastAPI, WebRTC (aiortc), PyTorch, YOLOv8, ByteTrack, Vision-Language Models, WebSockets, React, SQLite
 - **Principal Component Transform (PCT) for Multi-Band Imagery** [Code] | *Course Project, GNR 607: Satellite Image Processing* (Oct'25-Nov'25)
 - Implemented **PCA/PCT from scratch** on 11-band Landsat-8 imagery using manual mean-centering, covariance via matrix multiplication and component generation by eigenvector projection, without `sklearn.PCA` or `np.cov` anywhere in the implementation.
 - Built an interactive GUI to load multi-band GeoTIFFs, visualise all principal components, reconstruct from the top-k components and report per-PC variance explained alongside reconstruction MSE for each top-k choice, making the cost of compression explicit.
 - Reported reconstruction error (MSE) against the original bands for each top-k choice, showing the cost of compression.**Tech:** Python, NumPy, Tkinter, tiffio, scikit-image, Dimensionality Reduction

TECHNICAL SKILLS

- **AI & Agentic Systems:** LLM Integration, Multi-Agent Orchestration, ReAct Agents, Tool Calling, RAG, Model Context Protocol (MCP), Prompt Engineering, Schema-Constrained Outputs, LLM-as-Judge Evaluation, Context Compression, Token & Cost Budgeting, Vector Databases, AI Observability (Langfuse)
- **Machine Learning & Data:** PyTorch, NumPy, Pandas, Scikit-learn, OpenCV, Dask, Bayesian Inference, Clustering, Dimensionality Reduction, Uncertainty Calibration
- **Programming Languages:** Python, JavaScript, C, C++, Java, SQL
- **Backend:** FastAPI, Node.js, Express.js, Flask, REST APIs, Microservices, Async Processing, Server-Sent Events, JWT, OAuth, RBAC, Idempotency, Rate Limiting
- **Databases:** PostgreSQL, MongoDB, Redis, MySQL, SQLite | **ORM/ODM:** Mongoose
- **Geospatial:** GDAL, Rasterio, GeoPandas, Shapely, STAC, QGIS, Leaflet, COG, Remote Sensing
- **DevOps & Tools:** Docker, Docker Compose, Git, GitHub Actions, CI/CD, Linux, Jira, pytest, mypy, ruff, $\LaTeX$
- **Practices:** System Design, API Design, Code Reviews, Test-Driven Development, Agile Delivery

KEY SUBJECTS COMPLETED

- Machine Learning, Artificial Intelligence, Applied Machine Learning, Satellite Image Processing, Remote Sensing, GIS, Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks

SCHOLASTIC ACHIEVEMENTS

- Secured **GATE (CS)** with **98.94 Percentile** (Mar'25)
- **Finalist**, HackSVIT Hackathon (May'22)
- **Gold Medalist** across **150+** affiliated institutes, Diploma in Information Technology, **Gujarat Technological University (GTU)** (Jan'22)
- **Winner**, Hacktober Hacks Hackathon (Oct'21)
- **Winner**, Hackout'21 (Pre-made Projects) (Sep'21)
- Awarded **Best Student Award**, Diploma in IT, LJ Polytechnic (Nov'19)
- **Rajya Puraskar** (Governor of Gujarat), highest state-level honour of **The Bharat Scouts & Guides** (Feb'17)

POSITIONS OF RESPONSIBILITY

- **Teaching Assistant**, IIT Bombay, teaching **30+ postgraduate students** (Jul'25-Present)
- **Interview Coordinator**, IITB Placement Office: **2,100+ students**, **300+ member team**, **10+ companies**. (Oct'25-Dec'25)
- **Lead Organizer**, HackVGEC 2023, a **state-level hackathon** (Feb'23-Apr'23)
- **Lead Organizer**, TechXIT 2022, an **institute-level technical festival** (Sep'22-Oct'22)